

Women Empowerment Survey Analysis Pipeline

What each script does and how outputs flow

Input

Raw Data

Surveys.xlsx (5 sheets)
Synopsis V6 PDF

01_explore_surveys.py

Explore

Reads Excel sheets, identifies sections and respondent counts

Outputs:
outputs/exploration/
workbook_overview.csv

02_clean_and_code_surveys.py

Clean & Code

Normalises text; codes yes/no, gender, age, community, missing values

Outputs:
outputs/cleaned/
cleaned_analysis_responses.csv

03_multi_response_coding.py

Multi-Response

Splits comma-separated answers (crops, info channels, livestock, techniques)

Outputs:
outputs/multi-response/
multi_response_summary.csv

04_descriptive_results.py

Descriptive Stats

Frequency tables, percentages, assigns thesis themes

Outputs:
outputs/descriptive_results/
all_descriptive_results.csv

05_restructure_matrix_questions.py

Restructure Matrix

Extracts sub-items for Women Rep Q19, Q21, Q23 (scale / categorical responses)

Outputs:
outputs/matrix-corrected/
corrected_matrix_frequencies.csv

06_update_results_with_corrected_matrix.py

Merge Matrix

Replaces old matrix rows with corrected sub-items in final results

Outputs:
Updated descriptive results & summaries

07_thematic_coding.py

Thematic Coding

Codes 16 key open-ended questions into themes using keyword-based rules

Outputs:
outputs/thematic_coding/
thematic_frequencies.csv
thematic_coding_review.xlsx

08_merge_thematic_results.py

Merge Thematic

Adds thematic rows to final results; old rows kept as raw_open_ended

Outputs:
Updated results with thematic tables

09_update_synopsis_mapping.py

Synopsis Mapping

Maps questions to 4 objectives / hypotheses; updates top findings

Outputs:
outputs/synopsis_mapping/
objectives_with_top_findings.xlsx
TOP_FINDINGS_BY_OBJECTIVE.md

10_validate_results.py

Validate

Checks respondent counts, missing dominance, fragmentation, themes, count logic

Outputs:
outputs/validation/
validation_status_summary.csv
result_tables_for_manual_review.xlsx

Notes:

- 01-04: Produce the core cleaned data and frequency tables.
- 05-06: Fix Women Representative matrix questions (Q19, Q21, Q23).
- 07-08: Add thematic coding for 16 important open-ended questions.
- 09: Map all results to the four synopsis objectives/hypotheses.
 - 10: Runs last to validate everything.
- Launch GUI: ./run_gui.sh → <http://localhost:8501>

streamlit_app.py
report_generator.py

GUI & Reports

Interactive browser app,
Word / PDF reports, visualisations

Outputs:
gui_report_test.docx
gui_report_test.pdf
<http://localhost:8501>